

Linear Regression Analysis

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Linear Models

AIC

The Akaike Information Criterion (AIC) is a statistical measure used to compare competing models fitted to the same dataset. Rather than assessing a model in isolation, AIC evaluates the relative quality of multiple models by balancing goodness of fit with model complexity. In general, models with lower AIC values are preferred because they achieve a better fit while avoiding unnecessary complexity (Zajic, 2022).

The AIC is calculated using the following equation:

$$AIC = -2\ln(L) + 2k$$

where L represents the likelihood of the model and k represents the number of estimated parameters. The second term acts as a penalty for increasing model complexity, reducing the risk of overfitting by discouraging the inclusion of unnecessary predictors (Zajic, 2022).

AIC is widely used for model selection because it provides a simple and efficient method for comparing alternative statistical models, particularly when a quick comparison between candidate models is required (Zajic, 2022). However, it is important to recognise that AIC only measures the relative performance of the models being compared. Consequently, the model with the lowest AIC is considered the best among the candidate models, but this does not necessarily imply that it provides a good fit to the data overall. As highlighted by Zajic (2022), all of the candidate models could perform poorly despite one achieving the lowest AIC value. Therefore, AIC should be interpreted alongside additional evaluation metrics, such as Mean Absolute Percentage Error (MAPE), and diagnostic plots to provide a more comprehensive assessment of model performance.

The models

Model 1 - The best two-predictor linear model to predict Overall

For the Index of Multiple Deprivation (IMD) dataset, the linear regression model using Employment and Living as predictor variables is not the strongest two-predictor model for estimating Overall deprivation. Best subset regression identified Employment and Income as the superior predictor combination, achieving a higher coefficient of determination ($R^2 = 0.9490$) than the Employment and Living model ($R^2 = 0.9195$). This indicates that the Employment and Income model explains a greater proportion of the variation in Overall deprivation and therefore provides a better overall fit to the data.

Further evidence supporting this conclusion can be observed from the regression diagnostic plots. The Normal Q–Q plot for the Employment and Income model demonstrates that the residuals follow the theoretical normal distribution more closely than those of the Employment and Living model, suggesting improved normality of residuals. Additionally, the Scale–Location plot shows a more consistent spread of residuals across the fitted values, indicating greater homoscedasticity and fewer violations of the constant variance assumption.

The Residuals vs Fitted plots for both models display no obvious systematic pattern, suggesting that the assumption of a linear relationship between the predictor variables and the response variable is reasonable. Furthermore, the Residuals vs Leverage plots identify a small number of potentially influential observations that may warrant further investigation. These observations could represent genuine variation within the dataset or potential data quality issues and should therefore be examined before drawing final conclusions. As noted by Anand (2023), interpreting regression diagnostic plots involves an element of subjectivity, meaning that different analysts may identify violations differently.

Overall, the Employment and Income model is the preferred two-predictor model because it demonstrates stronger explanatory power, improved diagnostic performance and a higher coefficient of determination than the Employment and Living model.

The following R code was used to perform best subset regression and evaluate candidate two-predictor linear regression models.

```
library(tidyverse)

library(olsrr)

imd = read_csv('imd2025_individual.csv')

model = lm(Overall~ Employment + Health + Living + Crime + Education +
Income + Barriers, data = imd)

ols_step_best_subset(model, max_order = 2)

model = lm(Overall~Living+Employment, data=imd)

summary(model)
```

The results of the best subset regression and fitted linear regression model are presented below.

```
Best Subsets Regression ----- Model Index
Predictors ----- 1 Employment 2 Employment
Income ----- Subsets Regression Summary ----
-----
----- Adj. Pred
Model R-Square R-Square R-Square C(p) AIC SBIC SBC MSEF FPE HSP APC ---
-----
----- 1 0.8794
0.8785 0.8753 790.5524 647.1889 267.6975 655.8373 1009.9800 7.7673
0.0593 0.1243 2 0.9490 0.9482 0.9456 262.4509 535.5855 156.1529
547.1167 430.4520 3.3349 0.0255 0.0534 -----
-----
----- AIC: Akaike Information Criteria SBIC:
Sawa's Bayesian Information Criteria SBC: Schwarz Bayesian Criteria
MSEF: Estimated error of prediction, assuming multivariate normality
FPE: Final Prediction Error HSP: Hocking's Sp APC: Amemiya Prediction
Criteria
```

```
Call: lm(formula = Overall ~ Living + Employment, data = imd)
Residuals: Min 1Q Median 3Q Max -4.4904 -1.4744 -0.3127 1.2436 6.8641
Coefficients: Estimate Std. Error t value Pr(>|t|) (Intercept) -4.92015
0.86653 -5.678 8.59e-08 *** Living 0.16861 0.02104 8.013 5.80e-13 ***
Employment 171.14205 4.52524 37.819 < 2e-16 *** Signif. codes: 0 ' '
0.001 ' ' 0.01 ' ' 0.05 ' ' 0.1 ' ' 1 Residual standard error: 2.269 on
129 degrees of freedom Multiple R-squared: 0.9195, Adjusted R-squared:
0.9183 F-statistic: 736.8 on 2 and 129 DF, p-value: < 2.2e-16
```

The following R code was used to generate the diagnostic plots for both candidate regression models. These plots were subsequently used to assess the underlying assumptions of linear regression and compare overall model performance.

```
library(ggfortify)
```

```

model = lm(Overall~ Living + Employment, data = imd)

autoplot(model, data = imd)

model = lm(Overall~ Employment + Income, data = imd)

autoplot(model, data = imd)

```

Figure 1

Regression diagnostic plots for the Employment and Living two-predictor linear model. These plots were used to assess the assumptions of linear regression, including normality, homoscedasticity and influential observations

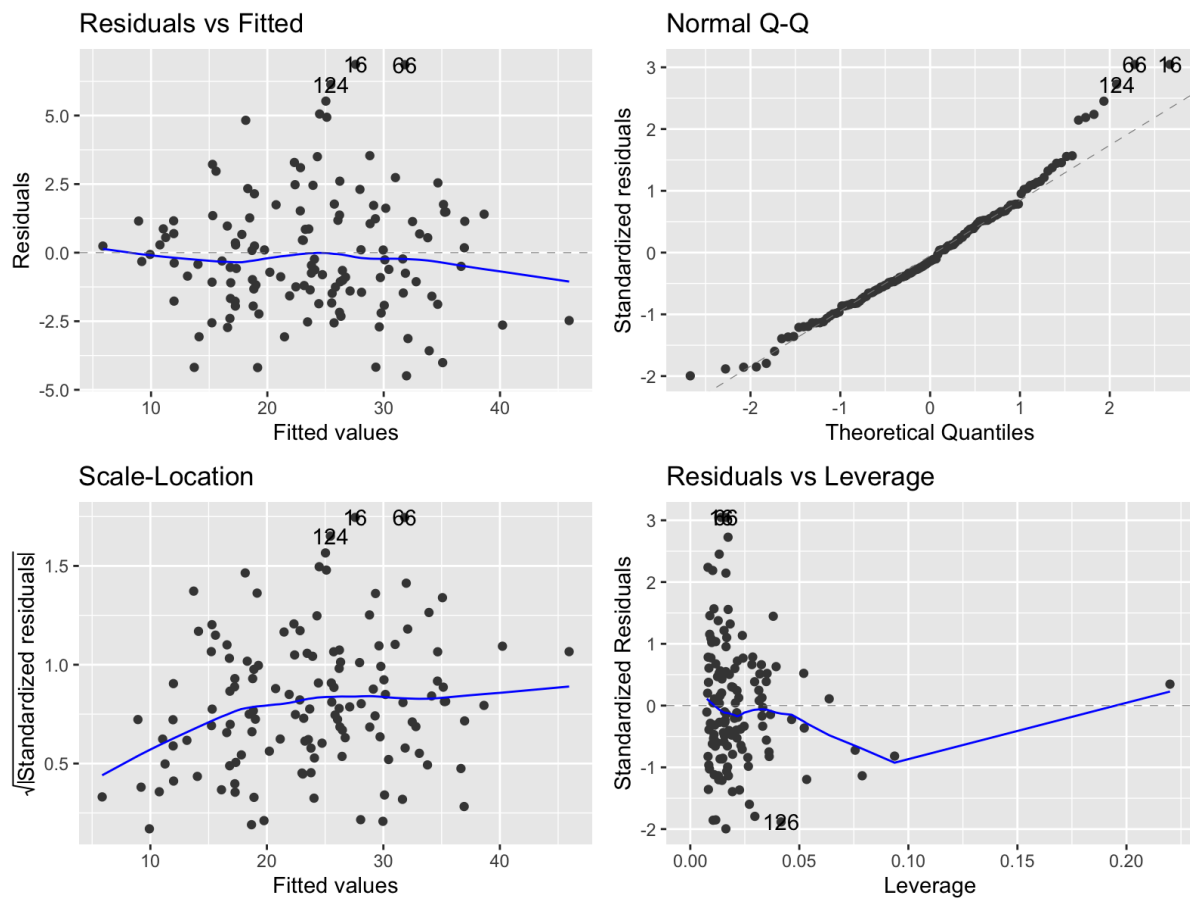
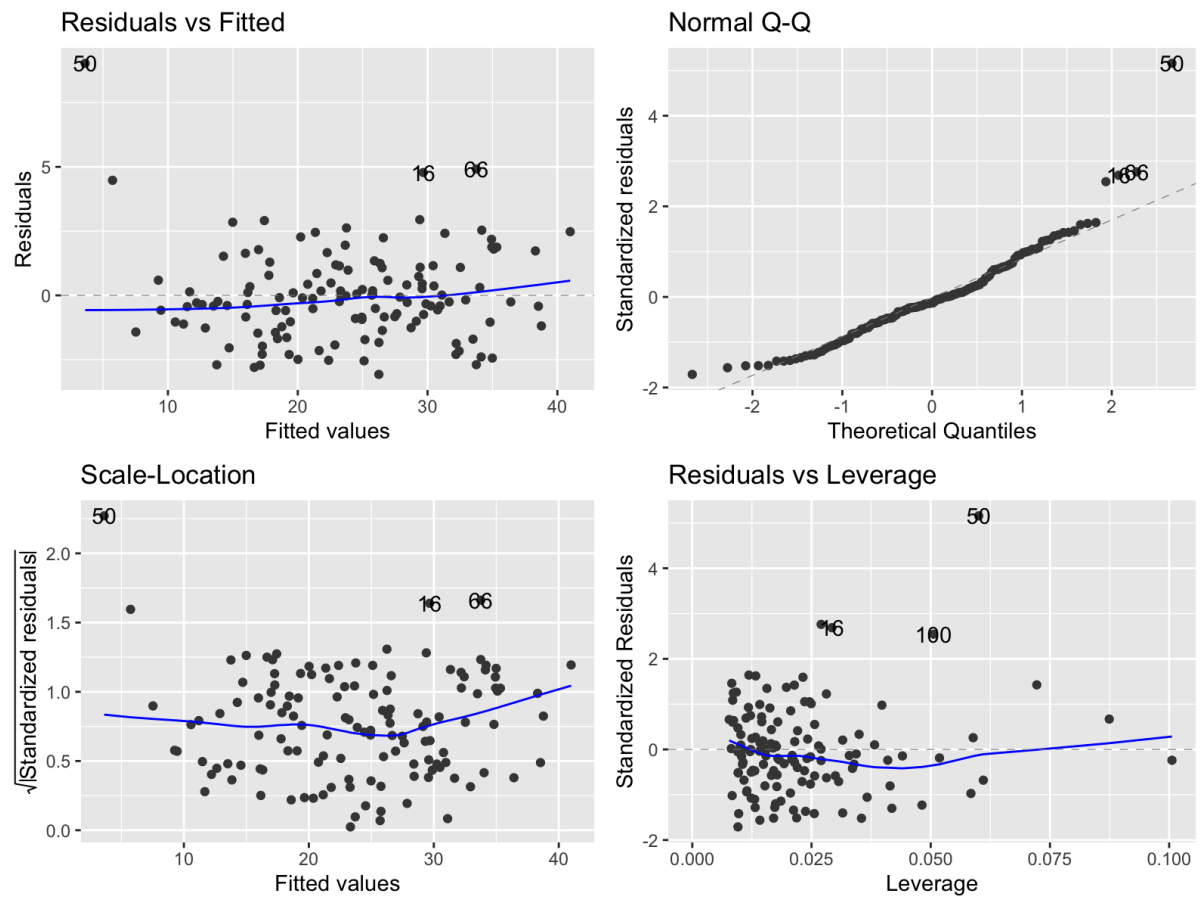


Figure 2

Regression diagnostic plots for the Employment and Income two-predictor linear model. Compared with Figure 1, these diagnostics indicate improved model performance and provide further evidence supporting the selection of this model



Model 2 - The best linear model to predict Overall that uses at most four quantitative predictors (excluding Rank)

To identify the strongest linear regression model containing a maximum of four quantitative predictors (excluding Rank), best subset regression was performed using all eligible deprivation variables. The objective was to determine the combination of predictors that provided the greatest explanatory power while maintaining a relatively simple and interpretable model.

The analysis identified Employment, Living, Education and Income as the optimal set of predictor variables. This model achieved an R^2 value of 0.9816, indicating that approximately 98% of the variation in Overall deprivation can be explained by these four predictors. Compared with the previous models, this represents a substantial improvement in explanatory power and demonstrates that incorporating additional relevant deprivation domains significantly improves model performance.

Furthermore, the selected model achieved the highest coefficient of determination while maintaining statistical significance for all predictor variables. This suggests that each variable contributes meaningful explanatory information and that the model effectively captures the underlying relationships within the dataset. Consequently, the combination of Employment, Living, Education and Income provides the most appropriate linear regression model within the specified constraint of using no more than four quantitative predictors

The following R code was used to perform best subset regression and identify the optimal four-predictor linear regression model.

```
model = lm(Overall~ Employment + Health + Living + Crime + Education +  
Income + Barrier,, data = imd)  
  
ols_step_best_subset(model, max_order = 4)
```

The results of the best subset regression analysis are presented below.

```
Best Subsets Regression -----  
-- Model Index Predictors -----  
---- 1 Employment 2 Employment Income 3 Employment Living Income 4  
Employment Living Education Income -----  
----- Subsets Regression Summary -----  
-----  
----- Adj. Pred Model R-Square R-Square R-  
Square C(p) AIC SBIC SBC MSEF FPE HSP APC -----  
-----  
----- 1 0.8794 0.8785 0.8753 790.5524  
647.1889 267.6975 655.8373 1009.9800 7.7673 0.0593 0.1243 2 0.9490  
0.9482 0.9456 262.4509 535.5855 156.1529 547.1167 430.4520 3.3349  
0.0255 0.0534 3 0.9626 0.9617 0.9588 161.1099 496.7589 117.2862  
511.1729 318.4249 2.4851 0.0190 0.0398 4 0.9816 0.9810 0.98 17.9818  
404.8591 29.6826 422.1559 157.5794 1.2388 0.0095 0.0198 -----  
-----  
----- AIC: Akaike Information  
Criteria SBIC: Sawa's Bayesian Information Criteria SBC: Schwarz
```

Bayesian Criteria MSEF: Estimated error of prediction, assuming multivariate normality FPE: Final Prediction Error HSP: Hocking's Sp APC: Amemiya Prediction Criteria

The regression summary demonstrates that all four predictor variables are statistically significant ($p < 0.001$), indicating that each contributes meaningfully to explaining variation in Overall deprivation. The model also produced a low residual standard error and an adjusted R^2 of 0.9810, suggesting an excellent fit while maintaining a relatively simple model structure.

Model 3 – The best linear model to predict Overall in London only, that uses at most four quantitative predictors (excluding Rank) but must include Crime

To investigate whether deprivation is influenced by different factors within London, a separate dataset containing only London districts was created. Best subset regression was then performed to identify the optimal linear regression model using a maximum of four quantitative predictors, while ensuring that Crime was included as a mandatory predictor. This allowed the influence of crime deprivation to be examined alongside the other deprivation domains.

The analysis identified Living, Crime, Education and Income as the optimal combination of predictor variables for predicting Overall deprivation within London. This model achieved an R^2 value of 0.9764, indicating that approximately 97.6% of the variation in Overall deprivation is explained by these four predictors. Furthermore, all predictor variables were statistically significant, demonstrating that each contributes meaningful explanatory information to the model.

Although the model explains slightly less variation than the best four-predictor model developed using the complete dataset, the results suggest that the factors influencing deprivation differ when the analysis is restricted to London. In particular, the inclusion of Crime reflects the additional importance of crime-related deprivation within the capital, illustrating how relationships between predictor variables can vary across different geographical regions. Consequently, the London-specific model provides a more appropriate representation of deprivation within this subset of the data than a model developed using all English districts.

To investigate deprivation within London specifically, a subset of the original IMD dataset containing only London districts was created. The following R code was then used to identify the optimal four-predictor linear regression model while requiring Crime to be included as a predictor variable.

```
London = read_csv('London.csv')

model = lm(Overall~ Employment + Health + Living + Crime + Education +
Income + Barriers, data = London)

ols_step_best_subset(model, include = c("Crime"), max_order = 4)
```

The results of the best subset regression analysis for the London dataset are presented below.

```
Best Subsets Regression -----
Model Index Predictors ----- 1
Crime 2 Crime Income 3 Living Crime Income 4 Living Crime Education
Income ----- Subsets Regression
Summary -----
-----
Adj. Pred Model R-Square R-Square R-Square C(p) AIC SBIC SBC MSEF FPE
HSP APC -----
----- 1
0.6656 0.6548 0.3773 674.1702 186.8386 87.5597 191.3282 493.3566
15.8543 0.4983 0.3775 2 0.9268 0.9219 0.8038 126.9773 138.7184 39.1198
144.7045 111.7585 3.6899 0.1166 0.0879 3 0.9628 0.9589 0.8523 53.3000
118.4019 19.4542 125.8845 58.8607 1.9950 0.0635 0.0475 4 0.9764 0.9730
0.8934 26.7143 105.4115 8.1734 114.3906 38.7560 1.3474 0.0433 0.0321 --
-----
----- AIC: Akaike
Information Criteria SBIC: Sawa's Bayesian Information Criteria SBC:
Schwarz Bayesian Criteria MSEF: Estimated error of prediction, assuming
multivariate normality FPE: Final Prediction Error HSP: Hocking's Sp
APC: Amemiya Prediction Criteria
```

Based on the best subset regression analysis, the optimal model for predicting Overall deprivation within London consists of the predictors Living, Crime, Education and Income. This model achieved an R^2 value of 0.9764, demonstrating excellent predictive performance while satisfying the requirement that Crime be included as one of the predictor variables.

To determine whether the same predictor variables were selected outside London, the analysis was repeated using a dataset containing all districts excluding London. The same best subset regression procedure was applied while again requiring Crime to remain within the model. This comparison enables an assessment of whether the factors associated with deprivation differ between London and the remainder of England.

```
Not_London = read_csv('Not_London.csv')

model = lm(Overall~ Employment + Health + Living + Crime + Education +
Income + Barriers, data = Not_London)

ols_step_best_subset(model, include = c("Crime"), max_order = 4)
```

The results of the best subset regression analysis for the non-London dataset are presented below.

```
Best Subsets Regression -----
Model Index Predictors ----- 1
Crime 2 Crime Income 3 Employment Crime Income 4 Employment Living
Crime Income ----- Subsets
Regression Summary -----
----- Adj. Pred Model R-Square R-Square R-Square C(p) AIC SBIC
SBC MSEF FPE HSP APC -----
----- 1 0.7610 0.7585 0.7441 1454.6428 562.3077 275.8608
570.0931 1630.9625 16.8071 0.1716 0.2489 2 0.9348 0.9334 0.9262
330.0096 435.7676 149.0486 446.1481 449.8940 4.6816 0.0478 0.0693 3
0.9581 0.9568 0.95 180.7136 393.9456 107.1025 406.9212 292.0566 3.0686
0.0314 0.0454 4 0.9802 0.9794 0.9768 39.1801 321.5657 38.2357 337.1364
139.2584 1.4772 0.0151 0.0219 -----
----- AIC: Akaike Information Criteria SBIC: Sawa's
Bayesian Information Criteria SBC: Schwarz Bayesian Criteria MSEF:
Estimated error of prediction, assuming multivariate normality FPE:
Final Prediction Error HSP: Hocking's Sp APC: Amemiya Prediction
Criteria
```

The optimal model for districts outside London consisted of Employment, Living, Crime and Income. While both models share Living, Crime and Income as important predictors, the London model selected Education, whereas the non-London model selected Employment. This suggests that although several deprivation domains consistently influence Overall deprivation, the relative importance of individual predictors varies between geographical regions. Consequently, separate regional models may provide a more accurate representation of deprivation than a single model fitted to all districts across England.

Considering the diagnostic plots produced by autoplot() discuss whether any of the districts in the dataset need further investigation

Of the three models developed, Model 2 was selected for further evaluation because it was fitted using the complete IMD dataset while also providing the strongest predictive performance. With an R^2 value of 0.9816, statistically significant predictor variables and a low residual standard error, this model provides the most representative description of the relationship between the deprivation domains and Overall deprivation. Consequently, its diagnostic plots were used to identify observations that may disproportionately influence the regression model.

Inspection of the diagnostic plots indicates that district 66 (Southend-on-Sea) and district 50 (Islington) warrant further investigation. District 66 exhibits large residuals across all four diagnostic plots, suggesting that the observed Overall deprivation differs substantially from the value predicted by the model. Such observations may indicate unusual characteristics within the district, data quality issues or limitations in the model's ability to explain that particular observation.

District 50 (Islington) demonstrates particularly high leverage in the Residuals vs Leverage plot. High-leverage observations have the potential to exert considerable influence on the estimated regression coefficients, meaning that including or excluding the observation could noticeably alter the fitted model. Consequently, influential observations should be investigated to determine whether they represent genuine variation within the data or potential anomalies requiring further attention.

Additional districts, including 129 (Merton), 89 and 92 (Brighton and Hove), also display unusual characteristics. Merton exhibits relatively high leverage, while Brighton and Hove display comparatively large residuals. However, these observations are less extreme than districts 50 and 66 and therefore appear less likely to substantially influence the overall regression model. Nevertheless, they remain worthy of further examination to confirm that their values accurately reflect the underlying data.

Further investigation of the dataset provides additional justification for examining these districts. Average deprivation scores calculated across each predictor revealed several combinations of deprivation measures that appear inconsistent with typical socioeconomic patterns. For example, district 50 exhibits higher-than-average Income deprivation alongside lower Education deprivation and approximately average Employment deprivation, while simultaneously displaying relatively high Living deprivation. Although these values may represent genuine socioeconomic complexity, they could also indicate unusual local circumstances or potential data quality issues. Similarly, district 129 exhibits comparatively low Education and Employment deprivation despite relatively high Living deprivation. These findings suggest that the identified observations should be investigated further to determine whether they reflect genuine regional variation, limitations of the regression model or possible inconsistencies within the dataset.

Overall, the diagnostic analysis demonstrates the importance of validating regression assumptions beyond summary statistics alone. Although Model 2 provides excellent predictive performance, the presence of influential observations highlights the need for additional investigation before drawing definitive conclusions from the model.

The following R code was used to generate the diagnostic plots for the selected regression model. These plots were subsequently used to assess the assumptions of linear regression and identify potentially influential observations.

```
model = lm(Overall~ Employment + Living + Education + Income, data = imd)
autoplot(model, data=imd, colour="Rank")
```

Figure 3

Regression diagnostic plots for the selected four-predictor linear regression model. The plots were used to evaluate model assumptions, including linearity, normality, homoscedasticity and the presence of influential observations.

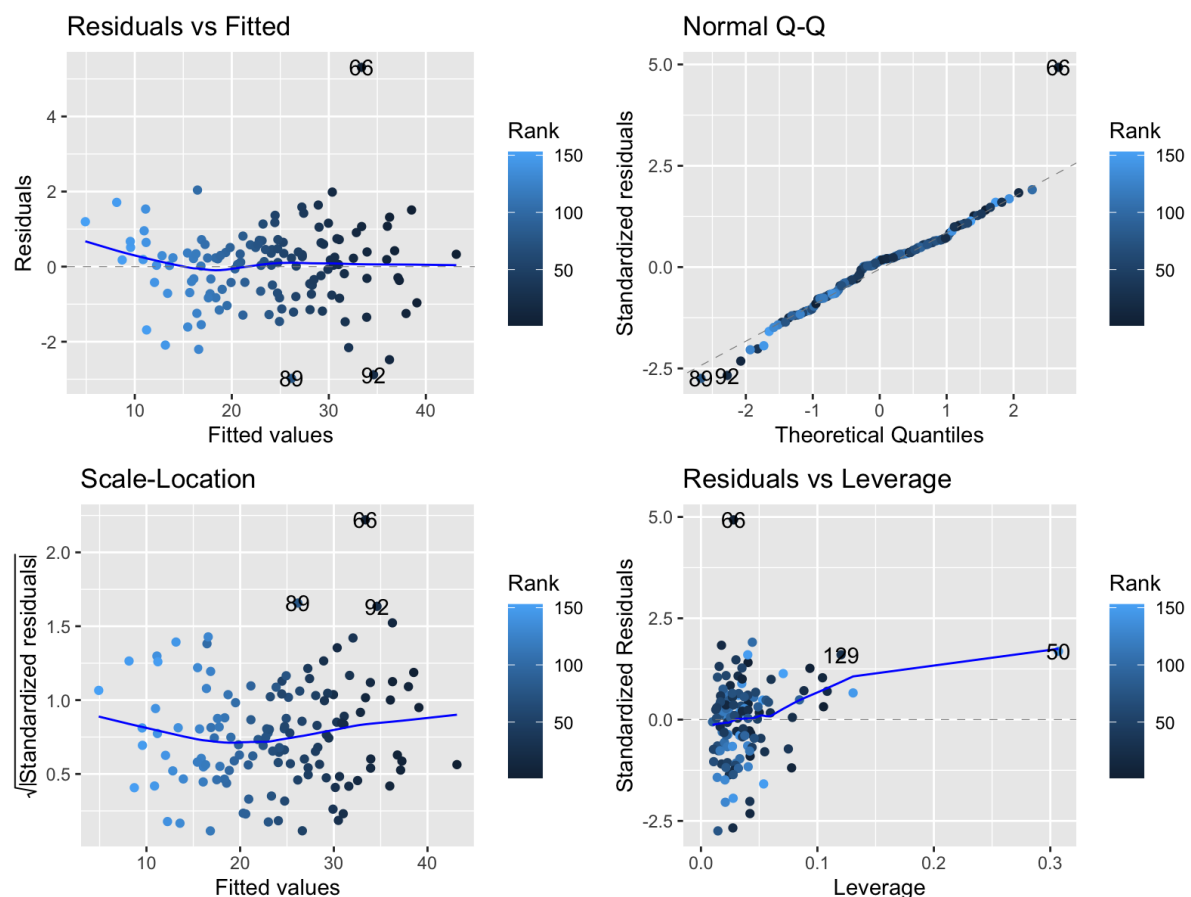


Table 1

Comparison of the three candidate linear regression models developed for predicting Overall deprivation.

Statistic	Model 1	Model 2	Model 3
Dataset	All districts	All districts	London only
Predictors	Living, Employment	Employment, Living, Education, Income	Living, Crime, Education, Income
R ²	0.9195	0.9816	0.9764
Adjusted R ²	0.9183	0.9810	0.9730
F-statistic	736.8	1696.0	289.1
Residual Standard Error	2.269	1.093	1.082
Median Residual	-0.3127	0.1774	0.1206
p-value	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$
AIC	595.87	404.86	105.41*
All predictors significant?	Yes	Yes	Yes
Overall assessment	Good two-predictor model, but outperformed by Models 2 and 3.	Best overall model for the complete IMD dataset. Highest explanatory power with excellent diagnostics.	Best model for London only; demonstrates that regional differences influence predictor importance.

* The AIC value for Model 3 is not directly comparable with Models 1 and 2 because it was fitted to a different dataset (London only).

The following R code was used to compare the performance of the three candidate linear regression models using a range of statistical evaluation metrics.

```
model = lm(Overall~Living+Employment, data=imd)

summary(model)

ols_step_best_subset(model, metric = "aic")

model = lm(Overall~ Employment + Living + Education + Income, data =
imd)

summary(model)

model = lm(Overall~ Living + Crime + Education + Income, data = London)

summary(model)
```

The comparison of the three candidate regression models is presented in Table 1.

```
Call: lm(formula = Overall ~ Living + Employment, data = imd)
Residuals: Min 1Q Median 3Q Max -4.4904 -1.4744 -0.3127 1.2436 6.8641
Coefficients: Estimate Std. Error t value Pr(>|t|) (Intercept) -4.92015
0.86653 -5.678 8.59e-08 *** Living 0.16861 0.02104 8.013 5.80e-13 ***
Employment 171.14205 4.52524 37.819 < 2e-16 *** Signif. codes: 0 '
0.001 ' 0.01 ' 0.05 '.' 0.1 ' 1 Residual standard error: 2.269 on
129 degrees of freedom Multiple R-squared: 0.9195, Adjusted R-squared:
0.9183 F-statistic: 736.8 on 2 and 129 DF, p-value: < 2.2e-16
```

```
Best Subsets Regression ----- Model Index
Predictors ----- 1 Employment 2 Living
Employment ----- Subsets Regression Summary
-----
----- Adj. Pred
Model R-Square R-Square R-Square C(p) AIC SBIC SBC MSEF FPE HSP APC ---
-----
----- 1 0.8794
0.8785 0.8753 65.2029 647.1889 271.1213 655.8373 1009.9800 7.7673
0.0593 0.1243 2 0.9195 0.9183 0.9164 3.0000 595.8703 221.4090 607.4015
679.6238 5.2654 0.0402 0.0842 -----
-----
----- AIC: Akaike Information Criteria SBIC: Sawa's
Bayesian Information Criteria SBC: Schwarz Bayesian Criteria MSEF:
Estimated error of prediction, assuming multivariate normality FPE:

Final Prediction Error HSP: Hocking's Sp APC: Amemiya Prediction
Criteria
```

```
Call: lm(formula = Overall ~ Employment + Living + Education + Income,
data = imd) Residuals: Min 1Q Median 3Q Max -2.9781 -0.6959 0.1774
0.5922 5.3087 Coefficients: Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.45896 0.42313 -12.90 <2e-16 *** Employment 76.80397
5.05544 15.19 <2e-16 *** Living 0.15789 0.01175 13.44 <2e-16 ***
Education 0.23970 0.02089 11.47 <2e-16 *** Income 35.76088 2.40949
14.84 <2e-16 *** Signif. codes: 0 ' ' 0.001 ' ' 0.01 ' ' 0.05 '.' 0.1 ' '
1 Residual standard error: 1.093 on 127 degrees of freedom Multiple R-
squared: 0.9816, Adjusted R-squared: 0.981 F-statistic: 1696 on 4 and
127 DF, p-value: < 2.2e-16
```

```
Call: lm(formula = Overall ~ Living + Crime + Education + Income, data
= London) Residuals: Min 1Q Median 3Q Max -2.0937 -0.6122 0.1206 0.5725
2.0062 Coefficients: Estimate Std. Error t value Pr(>|t|) (Intercept) -
3.72894 1.41405 -2.637 0.013492 * Living 0.20828 0.02718 7.663 2.39e-08
*** Crime 2.12541 0.85393 2.489 0.019026 * Education 0.26670 0.06647
4.012 0.000406 *** Income 55.77600 5.71282 9.763 1.63e-10 *** Signif.
codes: 0 ' ' 0.001 ' ' 0.01 ' ' 0.05 '.' 0.1 ' ' 1 Residual standard
error: 1.082 on 28 degrees of freedom Multiple R-squared: 0.9764,
Adjusted R-squared: 0.973 F-statistic: 289.1 on 4 and 28 DF, p-value: <
2.2e-16
```

This investigation demonstrated how different statistical modelling approaches can be used to understand complex socioeconomic relationships. While no single model is universally optimal, comparing multiple regression models highlighted how predictor importance can vary depending on the geographical scope of the analysis. The project also reinforced the importance of validating statistical assumptions through diagnostic plots rather than relying solely on goodness-of-fit measures. Overall, this investigation strengthened my understanding of regression modelling, model selection and the interpretation of statistical evidence within a real-world socioeconomic context.

References

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Zajic, A. (2022, November 29). *What Is Akaike Information Criterion (AIC)?* | Built In.

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